

BY 538: Biomedical Instrumentation and Diagnostic Technologies

Course Code	BY 538
Course Name	Biomedical Instrumentation and Diagnostic Technologies
L-T-P-C	3-0-0-3
Prerequisites	IC 136 Understanding Biotechnology & its applications or BE 309 Biosensing and Bioinstrumentation
Intended for	UG/PG/PhD
Mutual Exclusion	NA
Approval	42nd Senate (Jan 2024), in the M.Tech Biotechnology (Medical Biotechnology specialization) curriculum; 65th BoA (Aug 2026)

Preamble

This course provides an advanced understanding of biomedical instrumentation and modern diagnostic technologies used in health-care, biomedical research, and precision medicine. It introduces the principles of biomedical instrumentation, physiological signal acquisition, biomedical sensors, diagnostic systems, and emerging medical technologies, with emphasis on their design, operation, and clinical applications. Students will gain an understanding of conventional and next-generation diagnostic platforms, including electrochemical, optical, molecular, and point-of-care technologies, as well as wearable and digital health systems. The course also provides a brief introduction to selected emerging technologies and advanced concepts covered in the later part of the course, providing students with an overview of current and future directions in biomedical diagnostics and healthcare technology. Through this multidisciplinary approach, students from diverse academic backgrounds will develop the knowledge needed to understand, design, and contribute to the development and translation of next-generation healthcare technologies.

Course Modules

- Module 1: Instrumentation and Device Engineering:** Principles of biomedical instrumentation and physiological signal acquisition, biomedical sensors and transducers, semiconductor physics for biosensors, electrochemical interfaces, sources of noise and signal optimization, instrumentation performance analysis (accuracy, precision, sensitivity, selectivity, LOD, dynamic range, calibration), design considerations for biomedical devices. (8 Hours)
- Module 2: Advanced Diagnostic Technologies and Biosensing:** Electrochemical diagnostics (amperometric, potentiometric, impedimetric and voltammetric methods), optical diagnostics (SPR, fluorescence, Raman, SERS, and photoacoustic), photoelectrochemical biosensors, molecular diagnostics, immunodiagnostics; nanomaterial-enabled biosensing; microfluidics and lab-on-chip systems, multiplexed diagnostics, analytical performance evaluation, clinical validation, recent advances in infectious disease, cancer and precision diagnostics. (14 Hours)
- Module 3: Smart Biomedical Devices and Digital Healthcare:** Wearable, implantable and flexible biomedical devices, bio-electronics, MEMS and microfabrication for biomedical applications, Internet of Medical Things (IoMT), digital biomarkers, artificial intelligence and machine learning for diagnostic decision support, continuous physiological monitoring and future directions in personalized medicine. (12 Hours)
- Module 4: Translational Biomedical Engineering and Medical Device Development:** Clinical problem identification; engineering design methodology; prototype development; finite element analysis and computational modeling (overview); reliability testing; biocompatibility evaluation; design verification and validation; quality management systems (ISO 13485); regulatory pathways (FDA, CDSCO, CE); risk analysis (ISO 14971); intellectual property, technology transfer, entrepreneurship and commercialization; ethical issues in AI-enabled diagnostics; emerging trends in precision medicine. (8 Hours)

Text Books

- John G. Webster & Amit Nimunkar - *Medical Instrumentation: Application and Design*
- Joseph D. Bronzino and Donald R. Peterson - *The Biomedical Engineering Handbook* 4th Edition, CRC Press.

References

- Anthony P. F. Turner, Isao Karube and George S. Wilson - *Biosensors: Fundamentals and Applications*
- Lela Buckingham - *Molecular Diagnostics: Fundamentals, Methods and Clinical Applications*
- Edward Sazonov and Michael R. Neuman - *Wearable Sensors: Fundamentals, Implementation and Applications*
- Allen J. Bard and Larry R. Faulkner - *Electrochemical Methods: Fundamentals and Applications*
- Challa S. S. R. Kumar - *Nanotechnology Characterization Tools for Biosensing and Medical Diagnosis*